

Multitool LLMs with Microsoft AutoGen

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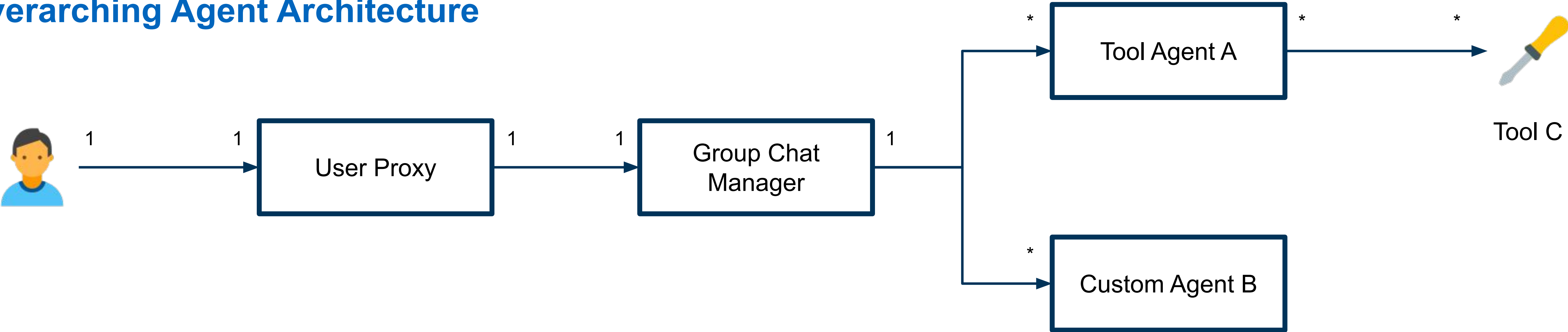
Overview

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- ❖ AutoGen is a Microsoft framework for building multi-agent LLM applications.
- ❖ Empirical studies demonstrate an advantage of the multi-agent setup over the traditional single-agent approach. This is more noticeable in specific task domains such as math, coding or RAG.
- ❖ As a part of this project we:
 - Conducted the experiments presented in the paper and reproduced the results
 - Explored additional scenarios to demonstrate the capabilities of the framework
 - Implemented a parent agent designed for an open-ended set of tasks and measured its performance
 - Developed a Streamlit demo to showcase the implemented scenarios

Overarching Agent Architecture

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Experiments

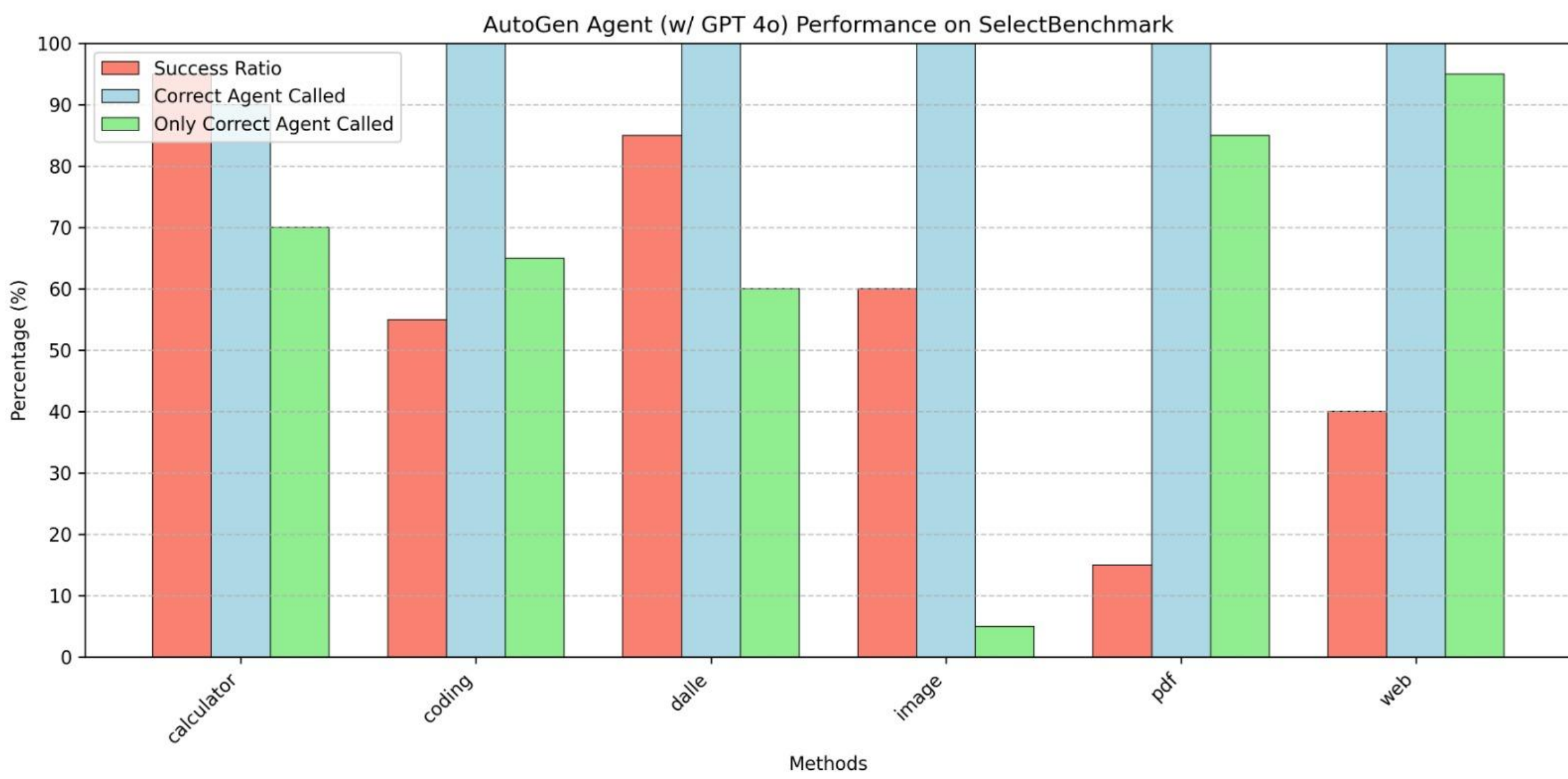
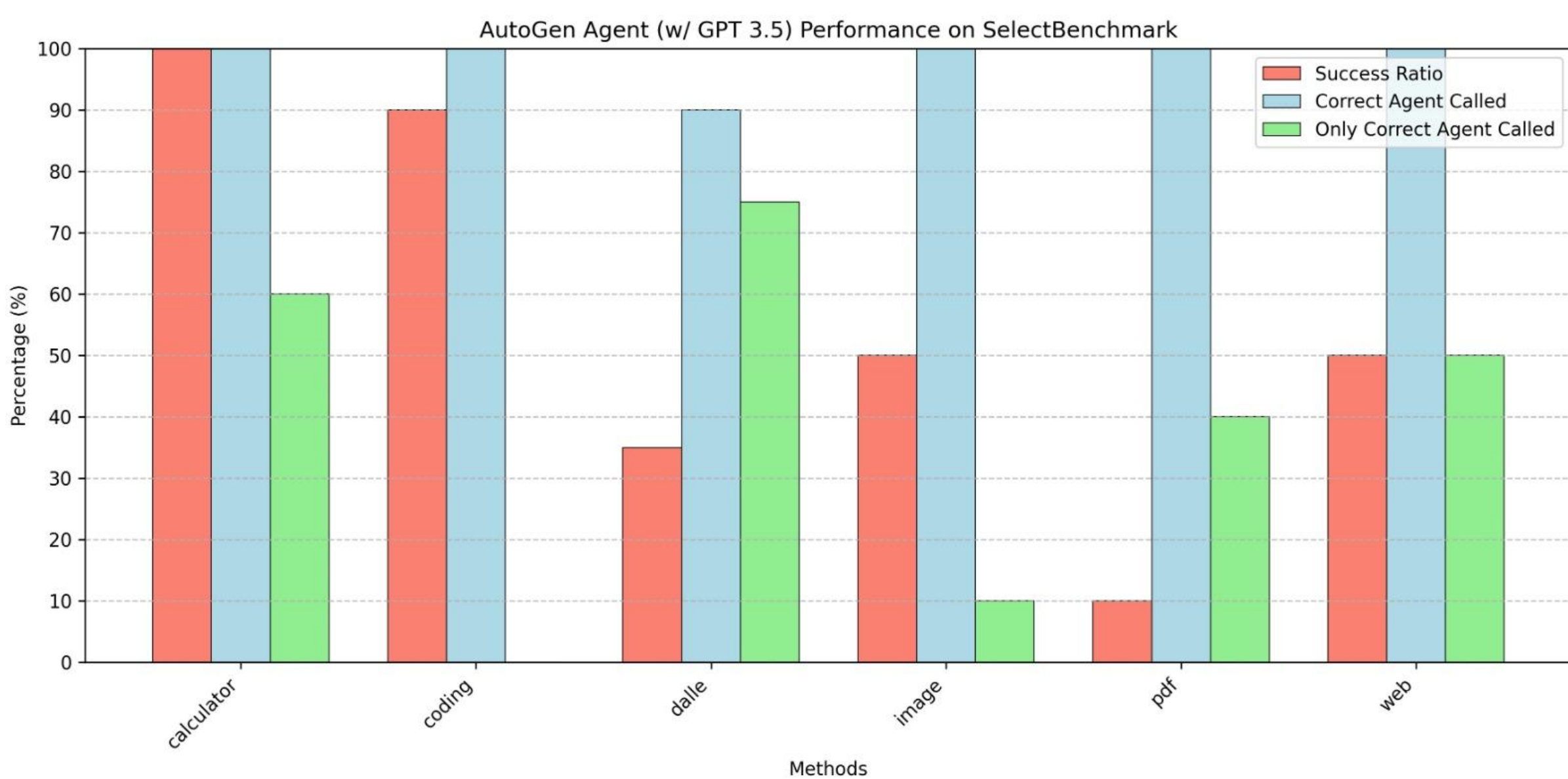
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- ❖ Reproduced the experiments conducted in the original paper
 - Some of the experiments showed similar results, others were more random
- ❖ Came up with further experiments to showcase the capabilities of the framework
 - PDFTriage: retrieving PDF context
 - DALL-E:
 - Critic agent improves user's prompt
 - Generator agent to create an image based on the improved prompt
 - Vision agent for image context
 - Browsing agent leveraging Google Search API and Selenium for direct site retrieval
- ❖ OpenAI models in general demonstrated better multiagent capabilities compared to Google Gemini

LLM-Based Evaluation

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- ❖ Developed the SelectBenchmark, a custom benchmark with 120 diverse tasks across 6 categories with 20 tasks each
- ❖ Benchmark goals:
 - Evaluate parent agent's ability to delegate tasks to the most appropriate child agent
 - Assess child agent's performance on the assigned task
- ❖ Automated evaluation using the LLM-as-evaluator approach for objective and scalable assessment
- ❖ Tested parent and child agents with both GPT-3.5 and GPT-4 backbones to compare performance



Overarching Agent in Details

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- ❖ Combines the functions of an agent set
- ❖ Delegates the task to the group chat manager, which in turn selects the agents from a given pool to help with the task
- ❖ Each agent in the diagram is a subset of agents solving a dedicated task
- ❖ Each agent can have a custom prompt and toolset - access to custom function calls

Streamlit Demo

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- ❖ Created Streamlit demos for
 - Standalone experiments
 - Overarching agent - with an option to add custom agents or select from a set of existing ones
 - LLM evaluation
- ❖ The demo is deployed to an Oracle Cloud VM via Docker

Limitations

- ❖ Multiplied price and time: each request generates several responses
- ❖ LLMs are unpredictable - a group of LLMs is even less so
- ❖ Development overhead

Future Work

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- ❖ Integrate more specialized agents into the flow
- ❖ Fine tune the agents on task-related datasets to provide better answers
- ❖ Test our parent agent against other similar frameworks